IB DP Psychology

Complete Cheat Sheet

Full Syllabus • HL + SL • Score a 7

Biological

Cognitive

Sociocultural

Research Methods

HL Extensions

Exam Mastery

20 Pages • IB Examiner-Level Content • Premium Edition

How to Use This Cheat Sheet • Syllabus Map

■ How to Use This Cheat Sheet

- Read each section systematically — Biological → Cognitive → Sociocultural
- Use colour-coded boxes: ■ Exam Tips (yellow) | ■■ Warnings (red) | ■/■ Evaluation (green/red)
- Study tables contain all key studies with Aim / Method / Results / Conclusion
- HL students: look for BLUE 'HL ONLY' banners for extension content
- Use Pages 16–20 the night before the exam for rapid revision
- Know your command terms: Describe, Explain, Discuss, Evaluate, To what extent

■ Full Syllabus Roadmap

Page	Section	Key Topics	Level
3–5	■ Biological Approach	Brain, neurons, hormones, neurotransmitters	SL + HL
6–8	■ Cognitive Approach	Memory, schema, decision making, thinking	SL + HL
9–11	■ Sociocultural Approach	Culture, SIT, norms, conformity, attribution	SL + HL
12–13	■ Research Methods	Experiments, ethics, reliability, validity	SL + HL
14–15	■ HL Extensions	Abnormal, Human Relationships, Health Psychology	HL only
16–20	■ Exam Mastery	Mistakes, strategy, quick revision, checklist	ALL

■ IB Exam Format at a Glance

- Paper 1: Three approaches (SAQ + ERQ) — 2 hours | SL + HL
- Paper 2: Options (ERQ) — 1 hour SL / 2 hours HL
- Paper 3 (HL only): Research methods — 1 hour
- Internal Assessment: Experimental study — 20% (SL) / 20% (HL)
- ERQ = Extended Response Question: typically 22 marks. SAQ = Short Answer Question: 9 marks each

Section 1 — Biological Approach: Core Concepts

■ Biological Approach — Pages 3–5

■ Key Brain Structures

- **Prefrontal Cortex — Decision-making, impulse contro**
- **Hippocampus — Memory consolidation (explicit/epis**
- **Amygdala — Emotion processing, fear response, agg**
- **Hypothalamus — Homeostasis, hormone regulation,**
- **Cerebellum — Motor coordination, procedural memo**
- **Broca's / Wernicke's area — Language production / c**
- **Corpus callosum — Connects left and right hemisphe**

■ Neurotransmitters

- **Dopamine — Reward, motivation, pleasure; linked to schizophrenia**
- **Serotonin — Mood regulation; low levels linked to depression**
- **Norepinephrine — Arousal, fight-or-flight**
- **Acetylcholine — Memory and muscle activation**
- **GABA — Inhibitory NT; reduces neuronal excitability**
- **Glutamate — Main excitatory NT; learning and memory**
- Synapse: NT released → crosses synaptic cleft → binds receptor

■ Neuroplasticity

- Brain's ability to change structure/function in response to experience
- **Synaptic pruning: unused connections eliminated (esp. adolescence)**
- **Long-term potentiation (LTP): repeated stimulation strengthens synapses**
- **Maguire et al. (2000): London taxi drivers had larger posterior hippocampi — evidence for experience-dependent plasticity**
- Critical/sensitive periods: windows when brain is especially malleable

■ Hormones & Behaviour

- **Cortisol — Stress hormone (HPA axis); high chronic levels impair memory**
- **Adrenaline (Epinephrine) — Fight-or-flight; enhances memory consolidation of emotional events**
- **Testosterone — Linked to aggression and dominance behaviour**
- **Oxytocin — Bonding, trust, social affiliation ('love hormone')**
- **Melatonin — Circadian rhythm regulation (sleep-wake cycle)**
- Hormones travel via bloodstream (slower but longer-lasting than NTs)

■ Exam Insight — Biological Approach

- SAQ: Always state the biological principle + the study + link to behaviour
- ERQ: Evaluate the approach — reductionism, determinism, animal studies
- Common question: 'Explain how one neurotransmitter influences behaviour'
- Use Maguire for neuroplasticity; use Caspi for gene-environment interaction

Biological Approach — Key Studies

■ Studies — Aim | Method | Results | Conclusion

Study	Aim	Method	Results / Conclusion
Maguire et al. (2000) London Taxi Drivers	Investigate structural brain differences in taxi drivers due to spatial navigation experience	MRI scans of 16 male taxi drivers vs 50 matched controls; voxel-based morphometry	Taxi drivers had significantly larger posterior hippocampi; positive correlation with years of experience
Caspi et al. (2003) 5-HTTLPR & Depression	Examine gene-environment interaction: 5-HTTLPR gene variant + stressful life events → depression	Longitudinal study; 847 participants followed from birth to age 26; measured serotonin gene and life stress	Individuals with short allele + high stress had significantly higher rates of depression — shows G×E interaction
Raine et al. (1997) Murderers' Brains	Identify brain abnormalities in murderers pleading NGRI (not guilty by reason of insanity)	PET scans of 41 murderers vs 41 matched controls; measured glucose metabolism in brain regions	Murderers showed reduced activity in prefrontal cortex, corpus callosum, parietal areas; supports biological basis of violence
Martinez & Kesner (1991) Acetylcholine & Memory	Investigate role of acetylcholine in spatial memory in rats	Rats injected with scopolamine (ACh blocker) or physostigmine (ACh enhancer); tested in maze	Scopolamine impaired maze performance; physostigmine improved it — ACh critical for memory formation
Baumgartner et al. (2008) Oxytocin & Trust	Investigate effect of oxytocin on trust-related behaviour after betrayal	RCT: participants inhaled oxytocin or placebo; played trust investment game with betrayal scenario	Oxytocin group showed significantly higher levels of maintained trust after betrayal vs placebo group

■ Localization of Function

- **Principle: specific brain areas are responsible for specific functions**
- **Broca's area (left frontal): speech production — damage = Broca's aphasia**
- **Wernicke's area (left temporal): language comprehension — damage = Wernicke's aphasia**
- **Case of HM (Scoville & Milner, 1957): removal of hippocampus → anterograde amnesia**
- Evaluation: fMRI studies support localization BUT brain is interconnected (holistic processing also occurs)

■ Study Tip — Biological

- Always include: Aim → Method → Results → Link to principle
- Caspi shows determinism is not absolute — G×E interaction is key for evaluation
- Maguire: quasi-experiment (no random allocation) — confound possible (self-selection bias)

Biological Approach — Evaluation

■■ Strengths, Limitations & Ethics

■ Strengths

- Scientific rigour: uses brain scans (MRI, PET, fMRI), experiments — high objectivity
- Real-world applications: drug treatments (SSRIs), brain surgery, neurological therapies
- Understanding of gene-environment interaction adds nuance (Caspi)
- Replicable: lab-based methods allow independent verification
- Cross-species research expands understanding (animal models)

■ Limitations

- Reductionist: reduces complex behaviour to biology alone, ignores cognitive/social factors
- Deterministic: implies genes/brain = destiny; undermines free will
- Animal studies (Martinez & Kesner): limited generalisability to humans
- Correlation ≠ causation (Maguire — did taxi driving cause hippocampal growth or did pre-existing trait attract them?)
- Technology-dependent: scans expensive, not widely accessible for clinical use

■■ Ethical Considerations

- **Informed consent: brain scan studies must ensure participants fully understand procedure**
- **Confidentiality: genetic information highly sensitive (Caspi — risk of discrimination)**
- **Harm protection: PET scans involve radioactive tracers — risk vs benefit analysis needed**
- **Animal welfare: Martinez & Kesner — rats injected with chemicals; 3Rs (Replace, Reduce, Refine)**
- **Dual-use problem: knowledge of biological bases of aggression could be misused**

■■ Common Mistakes — Biological Approach

- Confusing neurotransmitters with hormones (NTs = synaptic; hormones = bloodstream)
- Stating Maguire 'proves' neuroplasticity — quasi-experiment, causal claims limited
- Forgetting to evaluate — description alone = 4/9 on SAQ maximum
- Using animal studies without acknowledging generalisability issue

■ HL Extension — Genetics & Behaviour

- Genetic similarities: MZ twins (100% DNA) vs DZ (50%) — used in twin studies
- Heritability coefficient: proportion of variance in trait due to genes (0–1)
- Adoption studies: separate nature from nurture (e.g., Heston, 1966 — schizophrenia)
- GWAS (Genome-Wide Association Studies): identify gene variants linked to disorders
- IMPORTANT: Genes rarely act alone — always discuss G×E interaction

Section 2 — Cognitive Approach: Core Concepts

■ Cognitive Approach — Pages 6–8

■ Multi-Store Model of Memory (Atkinson & Shiffrin, 1968)

- Sensory Register → Short-Term Memory (STM) → Long-Term Memory (LTM)
- **Sensory Register: unlimited capacity, <1 sec duration, modality-specific**
- **STM: capacity ~7±2 items (Miller), duration 15–30 sec without rehearsal**
- **LTM: unlimited capacity, potentially permanent, semantic/episodic/procedural**
- Transfer: rehearsal (maintenance) moves info from STM → LTM
- Evaluation: oversimplified — HM showed different LTM stores; WM model more nuanced

■ Working Memory Model (Baddeley & Hitch, 19

- **Central Executive — attentional control, coordinates**
- **Phonological Loop — verbal/auditory info (inner spee**
- **Visuospatial Sketchpad — visual/spatial information**
- **Episodic Buffer (added 2000) — integrates informatio**
- Evidence: dual-task studies; amnesia patients (HM prese

■ Schema Theory (Bartlett, 1932)

- **Schema: cognitive framework/script that organises knowledge**
- Schemas influence encoding, storage, and retrieval
- Memory is reconstructive, not reproductive
- Bartlett: 'War of the Ghosts' — participants distorted story to fit cultural sc
- Evaluation: explains memory errors/false memories; difficult to define/mea

■ Thinking & Decision Making

- **Dual Process Theory (Kahneman): System 1 (fast, automatic) vs System 2 (slow, analytical)**
- **Cognitive biases: heuristics that lead to systematic errors in judgment**
- **Availability heuristic: judge probability by ease of recall (Tversky & Kahneman, 1973)**
- **Representativeness heuristic: judge probability by similarity to prototype**
- **Anchoring bias: over-reliance on first piece of information received**
- **Belief perseverance: maintaining beliefs despite contradicting evidence**

■ Exam Insight — Cognitive Approach

- MSM = good for explaining amnesia BUT Working Memory Model better explains multi-tasking
- Always evaluate models: What does the model NOT explain?
- Loftus & Palmer is the 'golden study' for cognitive — know all details
- Command term 'discuss' = strengths AND limitations + studies

Cognitive Approach — Key Studies

■ Studies — Aim | Method | Results | Conclusion

Study	Aim	Method	Results / Conclusion
Loftus & Palmer (1974) Leading Questions	Investigate how post-event information affects memory for a car accident	Lab experiment; 45 students watched car crash film; given questionnaire with verb manipulation ('smashed'/'hit'/'contacted')	Participants in 'smashed' condition gave higher speed estimates (40.8 mph vs 31.8 mph); in follow-up more reported broken glass (false memory)
Bartlett (1932) War of the Ghosts	Investigate how cultural schemas affect memory recall	Participants read unfamiliar Native American folk tale; recalled it after intervals (serial and repeated reproduction)	Participants omitted unfamiliar details, added familiar culturally consistent elements, distorted story — memory is reconstructive
Baddeley (1966) Acoustic vs Semantic Encoding	Investigate whether STM and LTM encode information differently	Participants given word lists (acoustically similar/dissimilar; semantically similar/dissimilar) — tested STM and LTM recall	STM: worse recall for acoustically similar words (acoustic encoding). LTM: worse recall for semantically similar words (semantic encoding)
Sharot et al. (2007) Flashbulb Memory & 9/11	Examine the neural basis of flashbulb memories using 9/11 as trigger event	fMRI scans of participants recalling 9/11 vs other summer events; compared amygdala activation	Downtown NYC participants showed greater amygdala activation and higher vividness/confidence — emotional proximity enhances memory encoding
Tversky & Kahneman (1973) Availability Heuristic	Investigate whether people use availability of examples to estimate frequency/probability	Participants asked whether there were more words starting with 'K' or with 'K' as 3rd letter; also assessed probability judgments	Most judged K-first as more common (wrong) — words starting with K are easier to retrieve; confirms availability heuristic

■ Cognitive Biases — Summary Table

- **Confirmation bias: seek/interpret info that confirms existing beliefs**
- **Overconfidence bias: overestimate accuracy of one's own knowledge**
- **Framing effect: decisions influenced by how options are presented**
- **Sunk cost fallacy: continue investment because of past costs (not future value)**
- **Hindsight bias: 'I knew it all along' — memory distorted after outcome known**

Cognitive Approach — Evaluation

■■ Strengths, Limitations & Applications

■ Strengths

- Scientific methods: lab experiments allow control of variables (Loftus & Palmer — high internal validity)
- Practical applications: eyewitness testimony reform, CBT (Cognitive Behavioural Therapy), education
- Models explain complex phenomena (WMM explains dual-task performance)
- Can be combined with biological approach (cognitive neuroscience, fMRI — Sharot et al.)
- Ecological validity improved by naturalistic studies and real-world triggers (Sharot — 9/11)

■ Limitations

- Artificial lab tasks may not reflect real-world cognitive processes (Loftus — film ≠ real accident)
- Computer metaphor (information processing) ignores emotion, motivation, social context
- Introspection and self-report issues: participants may not accurately report mental processes
- Individual differences ignored: schemas vary across people and cultures
- Models (MSM, WMM) are theoretical — brain structures don't map perfectly onto components

■ Real-World Applications

- **Eyewitness testimony: Loftus's research led to changed police interviewing practices (Cognitive Interview)**
- **CBT: based on idea that maladaptive cognitive schemas cause psychological disorders**
- **Education: schema theory informs curriculum design — build on prior knowledge**
- **User experience design: cognitive load theory shapes how apps/interfaces are built**
- **Legal system: understanding false memories has reformed how courts treat witness testimony**

■■ Common Mistakes — Cognitive Approach

- Confusing MSM and WMM — they are different models; WMM does NOT include LTM as a component
- Saying Loftus 'proved' memory is unreliable — she showed post-event info CAN distort memory
- Ignoring cultural context in schema theory — schemas are culturally constructed
- Not evaluating both sides — cognitive approach has real strengths, not just limitations

■ HL Extension — Cognitive Processing in the Digital World

- Screen time and attention: multitasking reduces depth of cognitive processing
- Cognitive offloading: using technology to reduce cognitive load (Google Effect — Sparrow et al., 2011)
- Filter bubbles: algorithms reinforce confirmation bias and limit exposure to diverse views
- Digital media and memory: auto-capturing events may reduce memory consolidation (Henkel, 2014)

Section 3 — Sociocultural Approach: Core Concepts

■ Sociocultural Approach — Pages 9–11

■ Culture & Behaviour

- Culture: shared beliefs, values, norms, practices transmitted across generations
- Individualism (Western): prioritises personal goals, independence, self-expression
- Collectivism (Eastern): prioritises group goals, interdependence, social harmony
- Hofstede's Dimensions: Power distance, Individualism, Uncertainty avoidance, Long-term orientation
- Enculturation: process by which individuals learn culture from birth
- Acculturation: process of adapting to a new culture (assimilation, integration, separation, marginalisation)

■ Social Identity Theory (Tajfel & Turner, 1979)

- Social categorisation: classify self and others into gr
- Social identification: adopt identity of group we belor
- Social comparison: compare in-group favourably to c
- In-group favouritism: prefer/benefit members of own
- Out-group discrimination: stereotype/disadvantage m
- Minimal group paradigm: even arbitrary groupings ca

■ Conformity & Norms

- Norms: shared rules about acceptable behaviour in a group or cultu
- Descriptive norms: what people typically do
- Injunctive norms: what people should/ought to do
- Informational conformity: conform because others seem to know be
- Normative conformity: conform to fit in/avoid rejection
- Asch (1951): 75% conformed at least once in line task — normative i

■■ Attribution Theory

- Attribution: explaining the causes of behaviour
- Internal (dispositional): behaviour caused by personal factors (personality, ability)
- External (situational): behaviour caused by context/environment
- Fundamental Attribution Error (FAE): overestimate dispositional, underestimate situational factors (Ross, 1977)
- Self-serving bias: attribute success to internal, failure to external causes
- Actor-observer bias: attribute own behaviour situationally, others' behaviour dispositionally
- Cultural variation: FAE is stronger in individualistic cultures (Miller, 1984)

■ Exam Insight — Sociocultural Approach

- SIT is one of the MOST tested topics — know the three-stage process
- Evaluation of SIT: explains prejudice, but ignores individual differences; does not explain all discrimination
- Always contextualise cultural studies: was sample WEIRD (Western, Educated, Industrialised, Rich, Democratic)?

Sociocultural Approach — Key Studies

■ Studies — Aim | Method | Results | Conclusion

Study	Aim	Method	Results / Conclusion
Tajfel (1971) Minimal Groups	Investigate whether arbitrary categorisation alone is sufficient to cause in-group favouritism	Boys categorised based on meaningless preference (Klee/Kandinsky); distributed points to in/out-group	Boys consistently awarded more points to in-group members even when anonymous and no prior interaction — categorisation alone causes discrimination
Asch (1951) Conformity	Investigate whether people conform to group norms when the answer is clearly wrong	Lab experiment; 123 US male participants; confederates gave obviously wrong answers on line-matching task	36.8% of critical trials participants gave wrong answer; 75% conformed at least once — normative social influence
Milgram (1963) Obedience to Authority	Investigate extent to which individuals obey authority even when causing harm	40 male US participants; instructed to deliver shocks to 'learner' (confederate); up to 450V	65% went to maximum voltage; obedience higher when authority figure present — situational factors override personal conscience
Berry (1967) Cultural Conformity	Investigate whether conformity differs between individualistic and collectivist cultures	Replicated Asch paradigm with Temne (collectivist, Sierra Leone) and Inuit (individualist, Canada)	Temne conformed significantly more than Inuit — supports cultural influence on conformity
Levine et al. (2005) In-group Favouritism & Football	Test SIT prediction that in-group identification affects helping behaviour	Manchester United fans; experimenter fell over wearing Man Utd, neutral, or rival team shirt	Participants most likely to help Man Utd shirt wearer — in-group favouritism mediates prosocial behaviour

■ Cultural Dimensions (Hofstede, 1980)

- Surveyed 116,000 IBM employees in 50+ countries on work-related values
- **Power Distance: acceptance of unequal distribution of power (high: Malaysia, low: Denmark)**
- **Uncertainty Avoidance: tolerance for ambiguity and uncertainty**
- **Masculinity-Femininity: competitiveness vs cooperation**
- Evaluation: Large sample + cross-cultural → generalisable; BUT corporate sample = not representative

Sociocultural Approach — Evaluation

■■ Strengths, Limitations & Applications

■ Strengths

- Recognises social and cultural context — more holistic than biological approach
- SIT has strong empirical support (Tajfel minimal groups; Levine et al.)
- Cross-cultural research (Berry; Hofstede) improves generalisability
- Practical applications: reducing prejudice, intergroup contact theory, multicultural education
- Explains collective behaviours (mob mentality, genocide, discrimination) that biology alone cannot

■ Limitations

- Oversimplifies culture — assumes cultures are homogeneous; ignores individual variation
- WEIRD bias: most studies conducted in Western/US contexts (Asch, Milgram)
- Asch: artificial lab task (line matching) — low ecological validity; may not reflect real conformity
- SIT: does not explain all intergroup conflict (e.g., economic competition — Realistic Conflict Theory)
- Ethical issues: Milgram — deception, psychological harm; Tajfel — induced discrimination

■ Real-World Applications

- **Reducing prejudice: Intergroup Contact Theory (Allport, 1954) — positive contact reduces out-group bias**
- **Multiculturalism policy: integration (Berry) shown to have best mental health outcomes**
- **Workplace diversity: SIT informs team-building and anti-discrimination training**
- **Media literacy: understanding attribution errors helps counter fake news and stereotyping**
- **Peacekeeping: understanding in-group/out-group dynamics informs conflict resolution strategies**

■■ Common Mistakes — Sociocultural Approach

- Confusing informational and normative conformity — they have different underlying processes
- Using Milgram without discussing ethical issues — always mention BPS/APA guidelines
- Stating culture 'determines' behaviour — it influences, not determines (avoid determinism)
- Ignoring individual differences within cultures — not all collectivists conform the same way

■ HL Extension — The Influence of Globalisation on Behaviour

- Globalisation → cultural convergence: adoption of Western values globally
- Research: Twenge et al. — rising individualism in China over decades
- Social media accelerates norm diffusion across cultures
- Threat to cultural identity → increased nationalism and cultural resistance

Section 4 — Research Methods: Core Concepts

■ Research Methods — Pages 12–13

■ Experimental Methods

- Lab experiment: high control, manipulate IV, measure
- Field experiment: natural setting; IV manipulated; hig
- Natural/quasi-experiment: IV naturally occurring (not
- IV (Independent Variable): what is manipulated by res
- DV (Dependent Variable): what is measured
- Extraneous variables: any variable that could affect D
- Confounding variables: EVs that systematically vary with IV

■ Research Designs

- Independent measures: different participants per condition (eliminat
- Repeated measures: same participants in all conditions (eliminates
- Matched pairs: pair participants on key variables, one per condition
- Longitudinal: same participants over extended time (tracks develop
- Cross-sectional: different age groups at same time point
- Case study: in-depth investigation of one individual (e.g., HM, Phine

■ Non-Experimental Methods

- Observation (naturalistic/controlled/participant): watch behaviour without intervention
- Interview (structured/semi-structured/unstructured): verbal data collection
- Questionnaire/Survey: self-report data; large samples; risk of response bias
- Correlational study: measure relationship between two variables; r = -1 to +1
- Content analysis: systematic analysis of text/media for themes/patterns
- Meta-analysis: statistical combination of results from multiple studies

■■ Research Ethics (BPS/APA Guidelines)

- Informed consent: participants must understand the study and agree to participate
- Deception: only permissible if essential AND followed by full debriefing
- Right to withdraw: participants can leave at any time without penalty
- Confidentiality: personal data must be protected and anonymised
- Protection from harm: physical, psychological, social harm must be minimised
- Debriefing: participants informed of true purpose after deception
- Animal research: 3Rs — Replace, Reduce, Refine (Home Office regulations)

■ Exam Insight — Research Methods (HL Paper 3)

- Paper 3 gives you an UNSEEN study to analyse — practise identifying: design, sampling, ethics, validity
- Know the difference between reliability (consistency) and validity (accuracy)
- Triangulation: using multiple methods increases credibility of findings
- Sampling: random, opportunity, purposive, snowball — know advantages of each

Research Methods — Data Analysis & Validity

■ Quantitative + Qualitative Analysis | Reliability & Validity

■ Quantitative Data Analysis

- **Descriptive statistics: mean, median, mode, range, st**
- **Mean: most sensitive; affected by outliers**
- **Standard deviation: spread of data around the mean**
- **Inferential statistics: test for significance (p < 0.05 = s**
- **Correlation coefficient (r): direction and strength of re**
- **Graphs: bar chart (categories), histogram (continuou**
- **Normal distribution: bell curve; most psychological tra**

■ Qualitative Data Analysis

- **Thematic analysis: identify recurring themes in textual data**
- **Content analysis: categorise and quantify qualitative content**
- **Grounded theory: generate theory from data inductively**
- **Reflexivity: researcher acknowledges own biases/influence on data**
- **Triangulation: compare multiple data sources/methods for consister**
- Rich, contextual data; cannot be generalised easily; subjective interpretat

Concept	Definition	How to Improve	Example
Internal Validity	Study measures what it claims to; no confounds	Standardise procedure; control EVs; use control group	Loftus: controlled setting; standardised film
External Validity	Findings generalise beyond study context	Use natural settings; diverse representative sample	Milgram: replicated in 8 countries
Ecological Validity	Tasks/setting reflect real-world behaviour	Naturalistic observation; field experiments	Levine: real helping behaviour in public
Test-Retest Reliability	Same results when repeated over time	Standardise procedure; use objective measures	Psychometric tests given twice to same group
Inter-rater Reliability	Agreement between different raters/observers	Operationalise categories; rater training; Cohen's Kappa	Observational studies coding same behaviour

■ Sampling Methods

- **Random sampling: every member of population has equal chance; most representative**
- **Opportunity/Convenience: use whoever is available; quick but potentially biased**
- **Purposive/Theoretical: select specific participants with characteristics of interest**
- **Snowball: participants recruit others; useful for hard-to-reach populations**
- **Stratified: divide population into strata, sample proportionally from each**
- IB IA: opportunity sampling most common; always discuss as a limitation

Section 5 — HL Extensions

HL ONLY — The following content is for Higher Level (HL) students only

■ Abnormal Psychology

Defining Abnormality

- **Statistical infrequency: abnormal = statistically rare (but: low IQ vs high IQ?)**
- **Deviation from social norms: violates cultural expectations (culturally relative)**
- **Failure to function adequately: unable to maintain daily life (Rosenhan & Seligman)**
- **Deviation from ideal mental health (Jahoda): absence of positive criteria**
- **DSM-5 (diagnostic manual): categorical system; 300+ disorders listed**
- **ICD-11 (WHO): alternative classification; wider international use**

Key Disorders — Summary

- **MDD (Major Depressive Disorder): persistent low mood, anhedonia, neurovegetative symptoms**
- **Schizophrenia: positive symptoms (hallucinations, delusions) + negative (flat affect, alogia)**
- **OCD: intrusive obsessions + compulsive rituals; serotonin and orbitofrontal cortex**
- **PTSD: flashbacks, avoidance, hyperarousal; trauma-induced; HPA axis dysregulation**
- **Anorexia Nervosa: distorted body image, restrictive eating; biopsychosocial model**
- Diagnosis issues: cultural bias in DSM; comorbidity; symptom overlap

❤■ Human Relationships

Formation, Maintenance & Dissolution of Relationships

- **Attraction: similarity (attitude alignment), proximity, reciprocal liking, physical attractiveness**
- **Social Exchange Theory (Thibaut & Kelley): relationships as cost-benefit analysis; CL and CLalt**
- **Equity Theory: satisfaction when perceived fairness in relationship inputs/outputs**
- **Attachment Theory (Bowlby): early caregiver bond shapes internal working models**
- **Duck's model of dissolution: intra-psychic → dyadic → social → grave-dressing phases**
- **Rusbult's Investment Model: commitment = satisfaction + investment – alternatives**

Prosocial Behaviour & Bystanderism

- **Bystander effect (Darley & Latané, 1968): less likely to help when others present**
- **Diffusion of responsibility: each bystander assumes someone else will act**
- **Pluralistic ignorance: assuming others' inaction signals no emergency**
- **Kitty Genovese case: 38 witnesses; led to founding of bystander research**
- **Empathy-altruism hypothesis (Batson): genuine altruism motivated by empathy**
- **Kin selection: evolutionary basis for helping relatives (Hamilton's rule: r × B > C)**

HL Extensions — Health Psychology

HL ONLY — Health Psychology — HL Only

■ Health Psychology

Stress & Health

- **Stress: imbalance between perceived demands and perceived coping resources**
- **HPA axis: hypothalamus → pituitary → adrenal cortex → cortisol release**
- **SAM axis: sympathetic → adrenal medulla → adrenaline/noradrenaline (fight-or-flight)**
- **Chronic stress: sustained cortisol → immune suppression, cardiovascular disease**
- **Kiecolt-Glaser et al. (1984): medical students' NK cell activity lower during exam period**
- **Lazarus & Folkman transactional model: primary appraisal (threat?) + secondary appraisal (can I cope?)**

Coping Strategies

- **Problem-focused coping: address the stressor directly (remove/change source)**
- **Emotion-focused coping: manage emotional response (meditation, social support)**
- **Avoidance coping: avoid/deny stressor — maladaptive long-term**
- **Social support: instrumental, emotional, informational — buffers stress**
- **Self-efficacy (Bandura): belief in own ability to cope — key protective factor**
- Individual and cultural differences in coping effectiveness

Health Behaviours — Promotion & Prevention

- **Health Belief Model (Rosenstock): perceived susceptibility + severity + benefits − barriers**
- **Theory of Planned Behaviour (Ajzen): attitude + subjective norm + perceived control → intention → behaviour**
- **Fear appeals: moderate fear most effective; must be paired with high efficacy message**
- **Motivational interviewing: non-judgmental approach to facilitate behaviour change**
- **Adherence to treatment: impacted by side effects, social support, patient-provider relationship**
- Tobacco, alcohol, exercise — key behaviours studied in IB Health Psychology

■ Strengths

- Directly applicable to reducing global health burden
- Integrates bio-psycho-social factors — comprehensive
- Informs public health campaigns and policy (smoking bans, vaccination messaging)
- Strong evidence base from randomised controlled trials

■ Limitations

- Self-report measures for health behaviours — subject to social desirability bias
- Difficult to establish causation (stress → illness) due to ethical constraints
- Cultural specificity of health models (HBM developed in Western context)
- Individual variation: same intervention works for some, not others

■ HL Key Studies — Health Psychology

- Kiecolt-Glaser et al. (1984): stress reduces immune function (NK cells) — natural experiment
- Ader & Cohen (1975): classical conditioning of immune response — psychological → physiological
- Bandura (1986): self-efficacy predicts health behaviour change

Exam Mastery — Common Mistakes

■■ Top Mistakes That Cost IB Psychology Students Marks

1. Describing, not evaluating
IB rewards evaluation heavily. Every major point needs: So what? What does this tell us? What are the limitations?

2. Confusing approaches
Don't mix up which study belongs to which approach. Maguire = biological, NOT cognitive.

3. Missing the command term
'Evaluate' ≠ 'describe'. Know: Describe (what) | Explain (why) | Discuss (+ and -) | Evaluate (judge critically)

4. Vague study references
Don't say 'a study found...'. Always: Author (Year) — Study Name — Key Finding.

5. Claiming causation from correlations
Correlation studies (Hofstede, Caspi) cannot establish cause. Say: 'there is an association between...'

6. Ignoring ethics
Ethics must be discussed for any study involving deception or harm (Milgram, Asch, Raine).

7. Forgetting to link to the question
Always explicitly link your evidence back to the question — examiners won't assume the connection.

8. One-sided ERQs
ERQs require balance — strengths AND limitations. One-sided answers cap at Grade 4.

9. Generic evaluation
'The sample was too small' is weak. Explain WHY size matters for THIS study.

10. Misusing 'proves'
Psychology cannot 'prove' anything. Use: suggests, indicates, provides evidence for, implies.

11. Ignoring cultural context
Always ask: where was this study done? WEIRD sample? Western findings may not generalise.

12. Confusing reliability and validity
Reliable = consistent results. Valid = measures what it claims. A study can be reliable but not valid.

13. Not defining key terms
Define every technical term first: 'Schema refers to a cognitive framework that...'

14. Over-describing the study
In ERQs, don't spend 70% of your response describing studies — evaluation and analysis are more valuable.

15. Sloppy IA analysis
In IA: state your findings, then link them back to the original study you are replicating.

Exam Strategy — SAQ & ERQ Mastery

■ How to Structure Your Answers

■ SAQ Structure (9 marks each — 3 per Paper 1)

- **~20–22 minutes per SAQ | Target: 250–350 words**
- **P — Point: State the principle/concept being addressed**
- **E — Evidence: Name the study (Author, Year), brief description of method**
- **R — Results: Key finding in specific terms (avoid vague language)**
- **E — Evaluate: One clear strength OR limitation of the evidence/approach**
- **L — Link: Explicitly connect back to the question**
- Marking: 1–3 marks (description) | 4–6 (with study) | 7–9 (with evaluation)

■ ERQ Structure (22 marks — 1 per Paper 1)

- **~45 minutes | Target: 800–1000 words | 3–5 paragraphs**
- **Introduction: Define key terms, outline your argument/position**
- **Body para 1–2: Present evidence WITH evaluation (not just description)**
- **Body para 3: Counter-argument or alternative explanation**
- **Conclusion: Balanced judgement — 'To what extent' questions NEED a verdict**
- Use: 'Furthermore...', 'However...', 'This is supported by...', 'In contrast...'
- Marking: Content (10) + Critical thinking (6) + Organisation (3) + Terminology (3)

Command Term	What It Requires	Marks
Describe	Give a detailed account of features/characteristics	SAQ
Explain	Give reasons/mechanisms; show understanding of 'why'	SAQ
Outline	Brief, structured account of main points	SAQ
Discuss	Present strengths AND limitations; balanced argument	ERQ
Evaluate	Judge value/importance; weigh evidence critically	ERQ
To what extent	Take a position; defend with evidence; acknowledge counter	ERQ
Contrast	Show clear differences between two things	ERQ

■ Golden Rules for IB Psychology Exams

- Never use 'I' in your answer — academic voice only
- Always name the researcher + year + study title
- One paragraph = one idea, one study, one evaluation point
- Counter-argument shows critical thinking — earns the top mark band
- Read the question TWICE before writing — mark the command term

Key Studies — Ultra-Condensed Master Reference

■ All Core Studies at a Glance

Study	Approach	Key Finding	Evaluation
Maguire et al. (2000)	Biological	Taxi drivers have larger posterior hippocampi — neuroplasticity	Quasi-exp; correlation only; male sample
Caspi et al. (2003)	Biological	5-HTTLPR short allele + stress → depression (G×E interaction)	Large longitudinal; self-report diagnosis; genetic reductionism
Raine et al. (1997)	Biological	Murderers show reduced prefrontal activity on PET scans	Small sample; all NGRI — not representative of all violence
Baumgartner et al. (2008)	Biological	Oxytocin maintains trust after betrayal vs placebo	RCT — high internal validity; lab setting reduces ecological validity
Loftus & Palmer (1974)	Cognitive	Verb manipulation ('smashed') inflates speed estimates; false memories created	Lab — high control; film ≠ real accident; student sample
Bartlett (1932)	Cognitive	Memory is reconstructive; schemas distort recall of unfamiliar stories	Low standardisation; lacks quantitative data; culturally outdated
Baddeley (1966)	Cognitive	STM = acoustic encoding; LTM = semantic encoding	Lab — artificial word lists; high internal, low ecological validity
Sharot et al. (2007)	Cognitive	Proximity to 9/11 increases amygdala activation & vividness of FBMs	fMRI = objective; self-report vividness = subjective; naturalistic
Tajfel (1971)	Sociocultural	Arbitrary categorisation causes in-group favouritism (minimal group)	Artificial task; male British boys only; demand characteristics
Asch (1951)	Sociocultural	36.8% conformity rate in line task — normative social influence	US male sample — WEIRD; lab task; deception used
Milgram (1963)	Sociocultural	65% delivered 450V shocks to authority figure's instruction	Ethical issues (deception, harm); lab artificiality; historical context
Berry (1967)	Sociocultural	Collectivist Temne conformed more than individualist Inuit — cultural variation	Cross-cultural strength; small samples; Asch paradigm criticism applies
Levine et al. (2005)	Sociocultural	Football fans help in-group shirt wearers significantly more	Field experiment — ecological validity; limited to sports context

Quick Revision — Bullet Summary of Everything

■ Rapid Fire Revision — Core Ideas Only

■ Biological — Key Points

- Brain structures: PFC (control), hippocampus (memory),
- NTs: dopamine (reward), serotonin (mood), GABA (inhibi
- Hormones: cortisol (stress), oxytocin (bonding), testoster
- Neuroplasticity: brain changes with experience (Maguire)
- G×E: genes + environment interact (Caspi — 5-HTTLPR)
- Ethics: animal studies, PET radiation, genetic privacy
- Evaluation: scientific but reductionist + determinist

■ Cognitive — Key Points

- MSM: Sensory → STM (7±2, 30s) → LTM (unlimited)
- WMM: CE + Phonological Loop + Visuospatial Sketchpad + Episodic Buff
- Schema: organises knowledge; memory is reconstructive (Bartlett)
- FBM: vivid emotional memories — amygdala role (Sharot)
- Cognitive biases: availability, anchoring, representativeness heuristics
- Loftus: post-event info creates false memories (leading questions)
- Application: Cognitive Interview; CBT; education design

■ Sociocultural — Key Points

- Culture: individualism vs collectivism (Hofstede)
- SIT: categorisation → identification → comparison → in-g
- Conformity: Asch (normative); Berry (cultural variation)
- Obedience: Milgram (65%; situational factors override mo
- Attribution: FAE; self-serving bias; actor-observer effect
- Prosocial: bystander effect; diffusion of responsibility (Da
- Evaluation: ecological validity issues; WEIRD samples; e

■ Research Methods — Key Points

- Designs: lab, field, natural experiments; case study; observation
- Variables: IV, DV, extraneous, confounding
- Ethics: consent, deception, withdrawal, confidentiality, harm, debrief
- Validity: internal (control), external (generalise), ecological (real-world)
- Reliability: test-retest; inter-rater (Cohen's Kappa)
- Data: quantitative (stats) vs qualitative (themes); triangulation
- Sampling: random (best), opportunity (common in IAs), snowball, stratifie

HL ONLY — HL Extensions — Quick Summary

Abnormal: Abnormality definitions — DSM-5 — MDD/Schizophrenia/OCD/PTSD | Human Relationships: SIT + attachment + bystander effect + prosocial behaviour | Health Psych: Stress (HPA/SAM) + coping + health belief models + adherence

■ Night Before: Final Scan List

- ■ Know 2–3 studies per approach with full AMRC
- ■ Practise one SAQ and one ERQ outline
- ■ Revise command terms — know the difference between 'discuss' and 'evaluate'
- ■ Review your IA — you will be asked to discuss it
- ■ Sleep well — cognitive function declines sharply with sleep deprivation!

Final Exam Checklist — Night Before & Exam Day

■ Your Complete Pre-Exam Checklist

KNOWLEDGE CHECKS

- ■ I can name 2–3 key studies per approach (Biological, Cognitive, Sociocultural)
- ■ I know each study's Aim, Method, Results, and Evaluation
- ■ I can explain all three approaches and their core principles
- ■ I understand Social Identity Theory and can apply it to a novel scenario
- ■ I know the Working Memory Model components and their functions
- ■ I can discuss neuroplasticity with reference to Maguire et al.
- ■ I understand G×E interaction and can explain Caspi et al.
- ■ I know the key ethical guidelines (BPS/APA) and can apply them

EXAM SKILLS CHECKS

- ■ I know the difference between SAQ and ERQ structure
- ■ I can identify all command terms and know what each requires
- ■ I always include a named study with author, year, and finding
- ■ My ERQs include both supporting evidence AND counter-argument
- ■ I define technical terms at the start of each answer
- ■ I link my evidence explicitly back to the question
- ■ I avoid words like 'proves' — I use 'suggests', 'indicates', 'provides evidence'

HL ADDITIONAL CHECKS

- ■ I understand all three HL options: Abnormal, Human Relationships, Health Psychology
- ■ For Paper 3: I can identify research design, sampling, and ethics in an unseen study
- ■ I know the definitions of abnormality and their cultural limitations
- ■ I can discuss the Health Belief Model and Theory of Planned Behaviour
- ■ I understand bystander effect and can apply it to novel scenarios

PRACTICAL LOGISTICS

- ■ Exam pens (x2 at minimum), pencils, ruler packed tonight
- ■ Exam timetable confirmed — know location and time
- ■ Student ID / exam number ready
- ■ Phone off and away BEFORE entering exam room
- ■ Sleep 7–9 hours — memory consolidation happens during sleep
- ■ Eat a proper breakfast — glucose fuels cognitive performance
- ■ Arrive 15 minutes early to settle nerves
- ■ If anxious: 4-7-8 breathing (inhale 4s, hold 7s, exhale 8s)

You've got this. Trust your preparation. Write clearly, evaluate critically, and manage your time.